

Cheng-Yuan Li

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, GPA: 4.0/4.0

Atlanta, GA

Aug. 2024 – May 2028

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, MySQL), C++, Bash, Java

Data Infrastructure: Google Cloud Platform (BigQuery, GCS), Terraform

Distributed Compute & Orchestration: Apache Spark (PySpark), Apache Airflow, Docker, dbt, Dataform

Libraries: Pandas, NumPy, Scikit-Learn

WORK EXPERIENCE

Data Engineer Intern

May 2026 – Aug 2026

Gordon Food Service, Inc.

Atlanta, GA

- Architected an automated cost-optimization engine using Python, Terraform, and Cloud Scheduler that dynamically allocates Dataform configurations between slots and on-demand pricing, capturing **\$50,000+ in annual infrastructure cost reductions**.
- Engineered an underlying metadata pipeline to ingest 3 TB of BigQuery audit logs, aggregating 4,000+ weekly execution records to feed the optimization engine while filtering pre-execution state queries to eliminate redundant processing overhead.
- Migrated a weekly 7 TB full-refresh production table to a Dataform incremental architecture; designed conditional merge operations via deterministic unique keys to **slash weekly data scan volume by 97.8%** down to 150 GB.

Teaching Assistant

Jan. 2025 – Present

Georgia Institute of Technology

Atlanta, GA

- **CS 4400 (Database Systems):** Instruct 100+ students on the end-to-end relational lifecycle, including ER/EER modeling, schema mapping, and normalization; led optimization of SQL joins, nested queries, stored procedures, and database indexing structures across a 4-phase project.
- **CS 1331 (Introduction to OOP):** Lead weekly code review labs and technical office hours for 220+ undergraduates, enforcing defensive programming, data structures, and test-driven development (TDD).

PROJECTS

NBA Analytics Data Pipeline | *Python, PySpark, PostgreSQL, dbt, Airflow*

Dec. 2025 – Present

- Architected a Medallion-pattern ETL pipeline using Airflow and Docker to ingest **9,100+** raw JSON files via the NBA API, implementing concurrency and exponential backoff to handle strict API rate limits seamlessly.
- Optimized storage footprints by designing an idempotent PostgreSQL star schema with conditional upserts, achieving a **10x storage reduction** over raw unstructured JSON formats.

SmallDB: High-Performance LSM-Tree Database | *C++*

Aug. 2025 – Present

- Designed and implemented an LSM-tree-based storage engine optimized for high write throughput, achieving **25%** faster sequential writes (5,748 ops/sec) and maintaining sub-200 μ s latency across diverse workloads.
- Engineered Bloom filter indexing that accelerated read operations by 1.8x (from 174 to 321 ops/sec), reducing read latency by **46%** while improving mixed read-write workload throughput by **73%**.
- Implemented a crash-consistent Write-Ahead Logging (WAL) subsystem with **100%** data durability, achieving near-instantaneous recovery at **33M+** ops/sec for baseline implementation.

RESEARCH

Research Intern

May 2025 – Aug. 2025

A*STAR Institute of High Performance Computing

Singapore

- Co-authored a model optimization research manuscript by engineering a profiling framework to benchmark 40+ transformer architectures, processing large-scale execution logs to map KV-cache footprints and hardware constraints against inference latency.